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SG21

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`noexcept(contract_assert(_))`

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Contradictory goals:

1. No technical reason to use `assert()`.
2. No technical reason to invent one's own assertion macro.
3. Contract annotations are for defining predicates, not for dodging contracts design choices.
4. Adding contract annotation does not affect surrounding compile-time semantics.
5. I want throwing violation handlers, because I know how to deal with exceptions.

Contract annotations are for defining predicates, not for dodging contracts design choices.

desired:

```
contract_assert(p(x));
```

not desired:

```
contract_assert {evaluate_in_build_mode_X, noexcept} (p(x));
```

Adding contract annotation does not affect surrounding compile-time semantics.

```
template <int I>  
Lib::X<I>::X(X&& r, int i = (contract_assert(I > 0), I)) noexcept;
```

```
std::is_nothrow_move_constructible<Lib::X<0>>
```

Not compiling controversial cases, reports the problem in the wrong place

library:

```
template <int I>  
Lib::X<I>::X(X&& r, int i = f(I)) noexcept;
```

programmer:

```
std::vector<Lib::X<0>> vec;  
vec.push_back({}); // ok
```

Not compiling controversial cases, reports the problem in the wrong place

library:

```
template <int I>  
Lib::X<I>::X(X&& r, int i = f((contract_assert(I > 0), I))) noexcept;
```

programmer:

```
std::vector<Lib::X<0>> vec;  
vec.push_back({}); // ERROR
```

Use cases for throwing violation handlers:

1. Detect bugs, but “never crash”.
2. Unit-testing defensive checks.
3. Try recovering, but OK to crash.

People who disable exceptions do not use noexcept.

Try recovering, but OK to crash.

- Works well today because we use different assertions than our libraries:
 - Our assertions throw – and we control them
 - Other assertions are disabled

“noexcept returns true, but we still throw”

- Not acceptable
- You get a throw, but it becomes different mechanism than normal exception handling
- Compromises expectation “use normal exceptions”.
- Exception handling means using noexcept for the strong guarantee.

Proposal

- Either remove throwing violation handlers
- Or accept that they are throwing, and say that